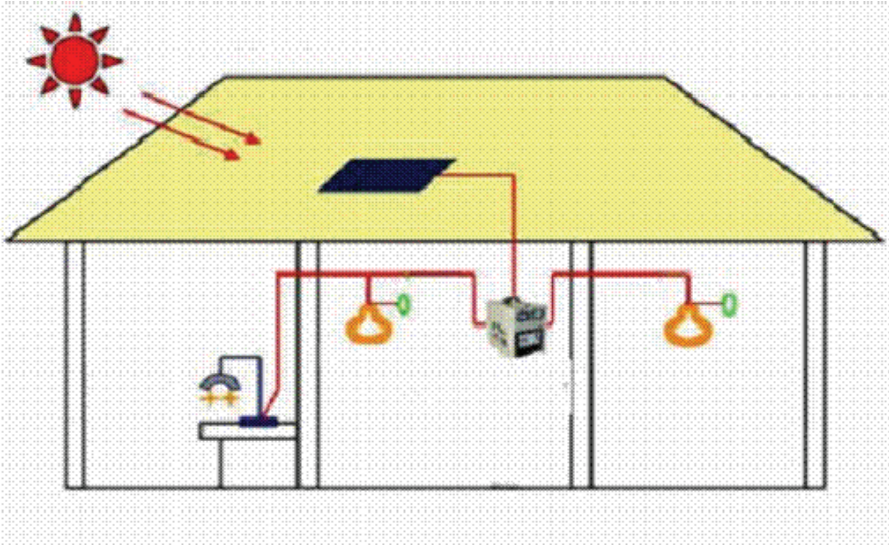
U-Tron (Beijing) Electronics Co., Ltd.

U-Tron
Solar Home Solutions



Environmentally Friendly

Economically Smart

www.u-tron.com

U-Tron®

About U-Tron (Beijing) Electronics Co., Ltd.

U-Tron (Beijing) Electronics Co. Ltd is a wholly owned American enterprise established in China in 1992. Our innovative technology is developed in the U.S and our core manufacturing is centralised in China.

As an engineering company, we possess the resources to design, develop and manufacture high power LED luminaries. We are committed to employing the latest technologies with the highest level of quality, service and value to our customers and suppliers through a process of continuous improvement.

Customers and suppliers are encouraged to visit our BDA factory to review our manufacturing process and the strict ISO 9000 quality standards imposed on our entire product range. Today, U-Tron owns 1000 square metres of office space located in the Central Business District of Beijing and a 1,800 square metre factory in the Beijing Development Area of Yi Zhuang District.

U-Tron's central goal is to promote the use of sustainable energy; a critical step in responding to our current deteriorating environment.






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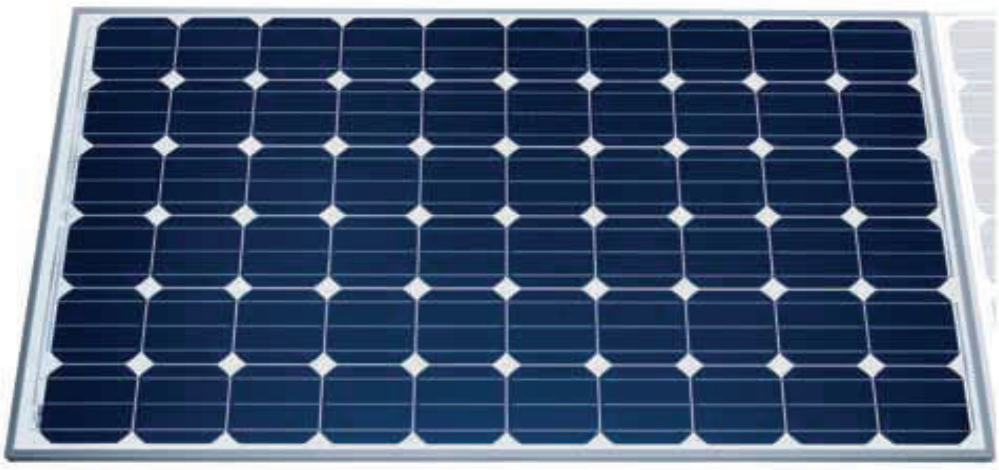
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U-Tron Solar Home System

Advantages:

- The system is almost free of charge after the initial investment, no more need to pay electricity fee
- Solar systems provide sustainable and green energy, so it helps to protect the environment
- An independent, movable lighting system with a high capacity, long life battery
- Our Helios LED Light Bulb with 300 Lumen is equivalent to a 40W electric incandescent lamp or a 18W energy saving CFL lamp.
- The system is very easy to install, so there's no need to hire an expert.



Applications:

Besides Home lighting, there are many other possibilities for using Solar LED Lighting Systems, Our system can be applied anywhere, such as remote areas, mountainous areas, desert areas, grassland areas, country areas. It can also be used for camping, outdoor activities and travel. The USB Connection can be applied to charge Mobiles, MP3, MP4, Cameras etc.

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Solar Home Package 1

Package 1 Part Number: A-6830-SPS-20W01



Contents	Individual Part Numbers
1 x 20W Solar Panel	6830-100-SP20
1 x 10A Controller	6830-121-C10
1 x 40 Ah Battery	6830-130-B40
3 x Helios LED Bulbs (3 W DC)	A-6850-P03-CR4E27-DC
3 x Switch	6830-190-E27HS
1 x Cabinet A with USB port	6830-150-C4U
1 x Cable (30 m twisted pair 1.5mm)	6830-160-C15
Installation Materials	

Recommended Application:

(Household up to 2 people)

- 1 LED bulb in the bedroom
- 1 LED bulb in the kitchen
- 1 LED bulb in the living room

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Solar Home Package 2

Package 2 Part Number: A-6830-SPS-40W02



Contents	Individual Part Numbers
2 x 20W Solar Panel	6830-100-SP20
1 x 10 A Controller	6830-121-C10
1 x 70 Ah Battery	6830-130-B70
4 x Helios LED Bulbs (3 W DC)	A-6850-P03-CR4E27-DC
1 x 10 W Fan (12 V DC)	6830-200-F12
4 x Switch + holders	6830-190-E27HS
1 x Cabinet B with USB port	6830-150-C6U
1 x Cable (30 m twisted pair 15mm)	6830-160-C15
Installation Materials	

Recommended Application: *(household up to 3 people)*

- 1 LED bulb in the bedroom
- 1 LED bulb in the kitchen
- 1 LED bulb in the living room
- 1 LED bulb in the bathroom

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Solar Home Package 3

Package 3 Part Number: A-6830-SPS-20W03



Contents	Individual Part Numbers
4 x 20 W Solar Panel	6830-100-SP20
1 x 20 A Controller	6830-121-C20
1 x 150 Ah Battery	6830-130-B150
5 x Helios LED Bulbs (3 W DC)	A-6850-P03-CR4B22-DC
5 x Switch + Holders	6830-190-E27HS
1 x Cabinet B with USB port	6830-150-C6U
1 x Cable (50 m twisted pair 15mm)	6830-160-C15
1 x Small Television (12 V DC)	6830-210-TV12
Installation Materials	

Recommended Application: *(Household up to 4 people)*

- 2 LED bulbs in two bedrooms
- 1 LED bulb in the kitchen
- 1 LED bulb in the living room
- 1 LED bulb in the bathroom

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Accessories/Add On's: Solar Panels

Technical Specifications	
Material	Mono/Polycrystaline Silicone
Max. Power	20/60 Watts
Dimension 20 Watts Panel	540mm*360mm*30mm
Dimension 60 Watts Panel	758mm*670mm*35mm
Weight	6 Kg
Optimum operating voltage (Vmp)	17.2V
Optimum operating current (Imp)	3.49A
Open-circuit voltage (Voc)	21.6V
Short-circuit current (lsc)	3.86A
Operating temperature	-40°C - 85°C
Maximum system voltage	600V DC
Power tolerance	± 3%

Solar Panel 20 Watts

This is the standard solar panel in all packages on pages 5, 6 & 7.

Part Number: 6830-100-SP20



Solar Panel 60 Watts

Instead of ordering three solar panels of 20 Watts each, you can also choose to order one 60 Watts solar panel.

Part Number: 6830-100-SP60



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Accessories/Add On's: LED Bulbs

Helios LED Bulbs Cool White:

U-Tron's ruggedized Helios Series LED light bulbs uses high power Cree LEDs that bring a healthy light to your home, while saving energy. They can be retrofitted easily to virtually any light fitting that uses E27, Bayonet or GU10 lamp holders.

Cool White is a bright color, especially fit for reading lights and other applications that desire full lighting.

Technical Specifications	
Power	3.5/5.5 Watts
Voltage	12V
Lumen 3.5V Bulb	390
Lumen 5.5V Bulb	520
CCT	5000 - 6000 K
Efficacy 3.5W Bulb	111.43
Efficacy 5.5W Bulb	94.5

Helios LED Bulb Cool White 3.5W

Part Number: A-6850-P03-CR4E27-DC



Helios LED Bulb Cool White 5.5W

Part Number: A-6850-P04-CR4E27-DC



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Accessories/Add On's: LED Bulbs

Helios LED Bulbs Warm White:

U-Tron's ruggedized Helios Series LED light bulbs uses high power Cree LEDs that bring a healthy light to your home, while saving energy. They can be retrofitted easily to virtually any light fitting that uses E27, Bayonet or GU10 lamp holders.

Warm White is a soft color, especially fit for atmospheric lights and other places that require gentle lighting.

Technical Specifications	
Power	3.5/5.5 Watts
Voltage	12V
Lumen 3.5W Bulb	321
Lumen 5.5W Bulb	428
CCT	2700 - 3500 K
Efficacy 3.5W Bulb	91.71
Efficacy 5.5W Bulb	77.82

Helios LED Bulb Warm White 3.5W

Part Number: A-6850-P03-WR4E27-DC



Helios LED Bulb Warm White 5.5W

Part Number: A-6850-P04-WR4E27-DC



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Accessories/Add On's: Batteries

Battery Construction:

1. Positive and negative plates consist of active mass and a lead-calcium alloy grid structure.
2. The safety valve opens when there is an abnormal increase in internal pressure caused by overcharging or misusage. Gas is released from the battery to return the pressure back to normal.
3. Container and covers are made of ABS or PP resin, with superior strength and acid resistance characteristics. The container and covers are sealed to prevent electrolyte and gas leakage.

Technical Specifications	
Nominal Capacity	40/70/150 Ah
Voltage	12V
Sealed Type	Sealed
Maintenance Type	Free

Battery I: 40 Ah

Part Number: 6830-130-B40



Battery II: 70 Ah

Part Number: 6830-130-B70



Battery III: 150 Ah

Part Number: 6830-130-B150



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Accessories/Add On's: Controllers

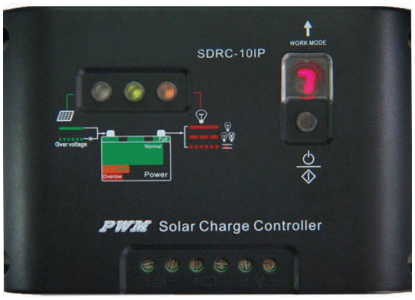
Solar Controllers Features:

1. 12/24V automatic recognition
2. High efficient Series PWM charging, which increases the battery life-time and improves the solar system performance.
3. Digital LED menu, only one key to adapt all settings simply.
4. Intelligent timer function with 1-15 hours option.
5. Gel, Sealed and Flooded battery type options.
6. Electronic protection against: overheating, overcharging, over dis-charging, overload and short circuit.

Technical Specifications	
Nominal System Voltage	12 / 24VDC Auto work
Battery Voltage Range	6-36V
Charge Circuit Voltage Drop	≤0.26V
Discharge Circuit Voltage Drop	≤0.15V
Self-consumption	≤6mA

Controller 10 Ampere

Part Number: 6830-121-C10



Controller 20 Ampere

Part Number: 6830-121-C20



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Accessories/Add On's: Controllers

Hybrid Controller Features:

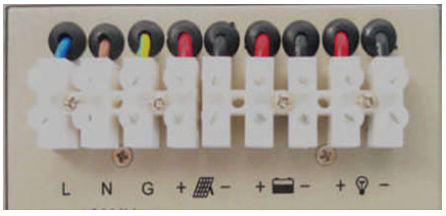
1. Microcontroller digital accuracy
2. PWM Charge mode
3. Specially designed for solar/utility hybrid system
4. Automatic detection of the voltage & full automatic operation
5. Over load protection

Technical Specifications:	
Rated Working Voltage	DC 12V
Output Voltage	DC 11.1-14.4V
Rated Charging Current	10A
Rated Load Current	5A
Max Output Power	60 Watts
AC Input Voltage	AC 170-264V
AC Input Frequency	47-63Hz
Grid Supply Voltage	DC 11.1V
PV Supply Voltage	DC 12.6V
Charging Mode	PWM

Hybrid Controller

Part Number: 6830-180-S0





Hybrid Controller

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Accessories/Add On's: Cabinets

Cabinets Features:

1. Design to in-house the controller and the battery.
2. Portable and lightweight, so it can be placed anywhere and moved whenever necessary.
3. USB Connection can be applied to charge Mobiles, MP3, MP4, Cameras etc.

Technical Specifications	
USB Outlet (Both Cabinets)	1 Outlet
DC Plug Outlet Cabinet A	3 Outlets
DC Plug outlet Cabinet B	6 Outlets
Size Cabinet A	285*180*260mm
Net Weight Cabinet A	4 Kg
Size Cabinet B	210*465*240mm
Net Weight Cabinet B	5 Kg
Color	Customized

Cabinet A with USB-port

Part Number: 6830-150-C4U



Cabinet B with USB-port

Part Number: 6830-150-C6U



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Installation & Maintenance

Installation:

1. According to the instructions for the line graph, solar panels should be installed in a sunny place. Plug solar panels into the socket according to the chart. The charging indicator light will flash when the solar panel is charging.
2. Connect the light plug according to the chart.
3. Turn on the main power switch and press the transmission button. The LED indicator light will turn on. Select large or small light and turn off the switch after each use.

Installation tips:

Here are three tips on dealing with the question on whether you should or shouldn't take a look at installing a system to generate solar energy for your home. By using these tips, you can potentially save thousands of dollars in energy costs during the year. Just imagine how much happier you and your family would be, as well as realizing how the home generation and use of solar power helps the environment.

Tip number one: Get knowledgeable on the topic.

If you just go and say, "hey, let's install some solar panels", you could be in for a rude awakening. What you need to realize is that there are basically 2 ways to go about it. The first one can be VERY expensive, and it calls for hiring a contractor to buy and install a complete solar energy system. This is a very expensive proposition which could cost in the tens of thousands of dollars, so find out if it is worth it in the long run with the money you will save. The second way is to do it yourself. This is not as problematic as it might seem, and there are several very good DYI solar energy guides available for fewer than fifty bucks. The most important thing here with either option is to do your research!

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Installation & Maintenance

Tip number two: Don't feel you have to get it all done at the same time.

One thing to keep in mind is that you do not need to run your whole house on solar power from the beginning, especially, as mentioned earlier, if you do not or cannot set up your whole place all at once. A do it yourself system can be installed to heat your hot water only. Depending on where you live, this can save you a bundle right there, and it is a project that can be done in a weekend.

With solar panels, the more you have, the more power can be stored to run things. After you have your hot water from this method, you can expand out for there at your own pace. It is a very doable thing.

Tip number three: Be creative and search for places in your home where you can start using solar energy first.

If you live in a house, look at all the things that use electricity. If you have a pool for example, or a hot tub, this is one of the perfect uses for the sun. Many newer units can be purchased with the solar heating and powering system already installed. If you have an older unit, a simple DIY solar panel system will work.

Your outdoor lighting system is another great candidate for solar power usage. Technology has advanced and solar batteries are great. One other thing, talking of batteries, do not forget buying solar powered electronics such as solar powered Bluetooth headsets, cell phones and MP3 players. Of course these tips are only touching the surface. Luckily, we live in the internet age where the correct information is just a simple search away. There is so much information out there on how to install solar energy for your home, and save tons of money as well as helping the environment at the same time.

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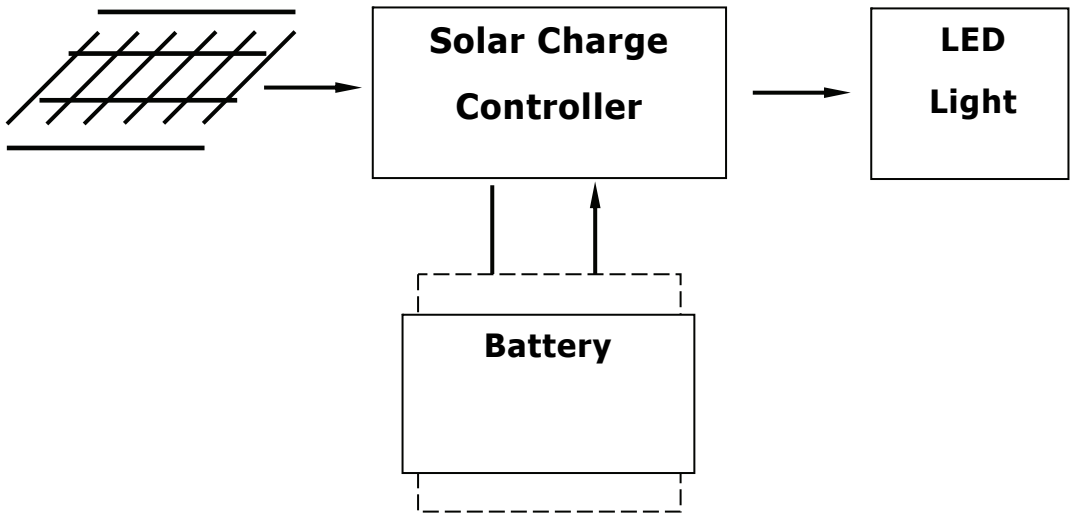
Installation & Maintenance

Maintenance

1. The solar panel should be installed to orientate the sun and fixed firmly. Make sure the solar panel clean and rotating insert the connector solidly
2. Please turn off the switch or use a small light to conserve electricity if people go out of the house.
3. Please turn off the switch after using to save the battery power. Observation the charge indicator and voltage meter timely and pay attention to charging status. Please use low-block lighting to save electricity When the power down and none enough.

Note:

1. The system is special. People must use our LED light and can not use other non-matching light. It's not allow to use other electrical connection, or it may cause equipment damage.
2. Due to the high power consumption for mobile phone charger, it is better to choose a clear day to charge your cell phone, and don't not to charge many phones in a day.
3. The power switch at the back-end of control box must be shut off if the system won't use for a long term.



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U-Tron Warranty

U-Tron (Beijing) Electronics Co. Ltd., warrant the products listed bellow for a period of 36 months from the date of shipment to be free from defects in material or workmanship and to be of the kind and quality designated or specified in the purchase agreements. If any products do not meet the above warranty and the buyer promptly files a warranty claim U-Tron shall do the following:

1. U-Tron shall provide a 12 month “No Questions Asked” replacement of any faulty units at no charge to the customer. The time limit for exchange shall not exceed 30 days from the time a claim is filed by the buyer. The buyer shall return the faulty units within 30 days of filing the claim. The method of returning the faulty units to U-Tron shall be decided at that time by U-Tron and all shipping charges shall be borne by U-Tron.

2. At the end of the 12 month period U-Tron shall provide additional free extended warranty for another 24 months. During this 24 month period U-Tron shall repair or replace and re-certify all faulty units returned. Shipping charges for sending the units to U-Tron shall be the responsibility of the Buyer and the Shipping charges to return the repaired goods to the buyer shall be borne by U-Tron. Time limit for repairs shall not exceed 60 days.

3. The warranties set forth herein shall be voided if the units have been 1) repaired or altered by any un-authorized person 2) subjected to misuse, negligence and accidents or operated in other than normal operation and use. 3) Connected, installed or modified in a manner not in accordance with U-Tron instructions.
   This warranty is applicable to all LED Products.

4. U-Tron at our own option can elect to provide replacement units at any desired percentage to be warehoused and logged by the buyers for timely service of the extended warranty replacements. When the warranty expires the bad units and the unused spare parts items shall be shipped back to U-Tron unless otherwise stipulated in the contract or purchase agreements. If the warehoused quantities become 90% used up or nearing the end of stock, the buyers shall ship back the faulty units to U-Tron. U-Tron shall rotate the spare parts inventory with in 60 days of receiving from the buyers.

5. All warranty claim items shall by shipped to the buyers regardless of the geographical location, by air freight to the nearest airport or courier service to the customer site, and any returning items can be shipped by sea or air freight at the request of U-Tron. If the customer fails to return the said faulty units with in the time frame as instructed, it will be considered as sold and U-Tron reserve the rights to bill the customer for its full value.

6. U-Tron shall not bare any responsibilities with the buyer’s customer claims or shall not bare any other expenses incurred other than the units cost it self. Customer site un-installation and re-installation of any faulty units shall not be the responsibility of U-Tron.

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U-Tron LED Lights

Save Energy,

Save the World!



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